C. Paprika and Permutation

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1300 | Tags: binary search, greedy, math, sortings | https://codeforces.com/problemset/problem/1617/C

Paprika loves permutations. She has an array $a_1, a_2, \ldots, a_n$. She wants to make the array a **permutation** of integers 1 to n.

In order to achieve this goal, she can perform operations on the array. In each operation she can choose two integers i ($1 \le i \le n$) and x (x gt 0), then perform $a_i := a_i \bmod x$ (that is, replace a_i by the remainder of a_i divided by x). In different operations, the chosen i and x **can be different**.

Determine the minimum number of operations needed to make the array a permutation of integers 1 to n. If it is impossible, output -1.

A permutation is an array consisting of n distinct integers from 1 to n in arbitrary order. For example, [2,3,1,5,4] is a permutation, but [1,2,2] is not a permutation (2 appears twice in the array) and [1,3,4] is also not a permutation (n=3 but there is 4 in the array).

Input

Each test contains multiple test cases. The first line contains a single integer t ($1 \le t \le 10^4$) — the number of test cases. Description of the test cases follows.

The first line of each test case contains an integer n ($1 \le n \le 10^5$).

The second line of each test case contains n integers $a_1, a_2, \ldots, a_n$. ($1 \le a_i \le 10^9$).

It is guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, output the minimum number of operations needed to make the array a permutation of integers 1 to n, or -1 if it is impossible.

Example

Input

```
4
2
1 7
3
1 5 4
4
12345678 87654321 20211218 23571113
9
1 2 3 4 18 19 5 6 7
```

Output

```
1
-1
4
2
```

Note

For the first test, the only possible sequence of operations which minimizes the number of operations is:

- Choose i=2, x=5. Perform $a_2 := a_2 \bmod 5 = 2$.

For the second test, it is impossible to obtain a permutation of integers from 1 to n.